

BRUNO C. M. BARRETO

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Software Development Engineer with expertise in data science, machine learning, system modelling, and dataset analysis seeking a challenging role as a Software Developer, Data Scientist, or Analyst, at an ambitious technology-driven company.

EDUCATION

Master of Computer Science – University of Illinois, Urbana-Champaign, IL	08/2026 – Ongoing
M.S. in Data Science - University of Washington, Seattle, WA – (3.95 GPA) (166 Verb. & Quant. GRE)	09/2023 – 04/2025
B.S. in Bioengineering with Data Science - University of Washington, Seattle, WA	09/2018 – 07/2022

EXPERIENCE

FULLSTACK SOFTWARE DEVELOPER - ZEZUME, SEATTLE, WA	05/2025 - Present
Developed an agentic AI full stack solution hosted in AWS for automated resume management and tailoring, using a serverless pattern with Cognito, API Gateway, Lambdas, DynamoDB, and S3 connected to a React single page application leveraging the Claude API to adapt resumes to employment targets.	
DATA SCIENTIST - VIRGINIA MASON, SEATTLE, WA	09/2024 - 03/2025
SPINE SURGERY COST FORECASTING MODEL - Developed a hybrid time-series forecasting model for Virginia Mason to predict U.S. spine surgery costs through 2030 based on operation, hospital, and patient data capable of explaining over 50% of the variance in historical surgery costs	
INFORMATICS SPECIALIST - NANOSTRING, SEATTLE, WA	12/2021 - 06/2022
GEOMX SUSTAINABLE REAGENT PROJECT - Reduced operating costs for the GeoMx Digital Spatial Profiler by altering reagent container and modifying device software to intelligently monitor fluid requirements, resulting in a 20% drop in reagent costs.	

CERTIFICATIONS

Azure Data Science Associate (DP-100)	09/2022
Azure Fundamentals (AZ-900)	07/2022
Data Science - General Assembly	02/2023
Azure AI Engineer Associate (AI-102)	08/2023
AWS Cloud Practitioner	06/2025

DATA SCIENCE PROJECTS

DEEP LEARNING IMAGE CLASSIFIER	04/2022 - 06/2022
Developed a model in Python to automatically classify images into 10 distinct categories using a trained convolutional neural network with 90% accuracy in classification.	
PREDICTIVE NERVE REGNERATION MODEL	05/2021 - 06/2021
Developed a model of peripheral nerve regeneration for neuroscientists that reduced growth factor conduit testing times by 100x using COMSOL Multiphysics.	
HOUSING PRICE ESTIMATOR	11/2022 - 12/2022
Created a machine learning model to automatically assign appropriate house prices for realtors in Ames, Iowa using a linear regression trained on local tax data, resulting in a model capable of accounting for 92% of variance in house prices.	
ACCIDENT SEVERITY PREDICTOR NLP	01/2023 - 02/2023
Developed a model that can predict flight accident severity from a formal report with 90% accuracy and determine that improper installation and maintenance of airframe components was a key cause of high lethality	
ATTENTION-BASED SENTIMENT CLASSIFIER	05/2024 – 06/2024
Developed an attention-based model to automatically determine the sentiment of a movie review from contextless review text with 88% accuracy	
TURBO GLYCOLYSIS PID CONTROLLER	04/2022 - 06/2022
Created a PID controller in Python and MATLAB for ATP production in an unstable turbo glycolysis bioreactor that improved reactor settling times by 80% and made reactions stable.	

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, MATLAB, R, C++, JavaScript, TypeScript

Database Languages: SQL, MySQL, PostgreSQL, Microsoft SQL Server, MongoDB, GraphQL, SparkSQL

Machine Learning Frameworks: Tensorflow, PyTorch, NumPy, scikit-learn, Pandas, OpenCV, spaCy, skforecast, Scipy, Statsmodels, Polars

Machine Learning Types: Regression, Classification, Natural Language Processing, Neural Networks, Time Series

Software Technologies: Git, Github, Slack, Jupyter Notebooks, Google Colab, Microsoft Office, Linux, Matplotlib, Seaborn, Tableau, Power BI, Flask, Kubernetes, Docker, Artificial Intelligence, Claude Code CLI, Claude API

Cloud Technologies: AWS, IAM, S3, Cognito, Lambda, API Gateway, DynamoDB, S3, EC2, Athena, Redshift, SageMaker, CloudWatch, Microsoft Azure, Azure AD, Azure VM, Azure Functions, Azure Blob Storage, Azure Files, Azure Synapse

WORK AUTHORIZATION

U.S. Citizen, authorized to work for any employer.